



AUT

TE WĀNANGA ARONUI
O TĀMAKI MAKAU RAU

Dentinal Hypersensitivity and Desensitising Agents

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Learning Objectives

- Define dentinal hypersensitivity and identify factors contributing to dentinal hypersensitivity
- Define and discuss Classic recession, Stillman's cleft and McCall's festoon
- Define dehiscence and fenestration
- Discuss the diagnosis and management of gingival recession, including classification
- Explain Brännström's hydrodynamic theory of pain transmission
- Discuss and cite examples of tactile, chemical, thermal, osmotic, and bacterial pain stimuli
- Identify four modes of action for desensitising agents
- Identify the active ingredients for currently marketed professional and patient-applied desensitising agents

Definition

- **Dentinal Hypersensitivity:**

A short, sharp pain arising from exposed dentine in response to thermal, tactile, osmotic, or chemical stimuli that cannot be explained by any other dental pathology.

(Addy et al., 1985; Canadian Advisory Board, 2003)



Dentinal Hypersensitivity

- **Aetiology** depends on partial or total loss of enamel (e.g., from attrition, abrasion, or erosive processes and their combinations)
- Dentinal tubules **must be open** to allow for fluid diffusion to cause hypersensitivity



Prevalence

- Several studies reported that between **14-100%** of the adult population suffers from dentine hypersensitivity
- All teeth can be affected but predominantly canines and premolars
- 50% of all persons affected are under 30 years
- Approx. 75% of patients experience sensitivity to cold stimuli
- Approx 30% to mechanical stimuli



Contributing Factors

1. Classic recession
2. Gingival and periodontal attachment loss
 - Gingival and periodontal diseases
 - Gingival recession
 - Due to orthodontic movement
3. Non-surgical / surgical periodontal therapy
 - resulting in the resolution of inflammation
 - shrinkage of oedematous tissue
4. Over-vigorous or incorrect tooth brushing pressure
 - dependent on biotypes



Contributing Factors cont.

5. Use of abrasive toothpastes
6. Anatomy of CEJ
7. Thin cementum
 - Readily removed during surface debridement & polishing
8. Chemical reaction
 - Associated with plaque, bulimia and diet



Contributing Factors cont.

9. Attrition

10. Abfraction??

- the dynamic stress that occurs during mastication or bruxism creates a cyclic tension and compression in cervical areas

11. Restorative procedures / oral surgery

12. Parafunctional habits

13. Root caries



Classic Recession

- **Defined:** An inflammation-free clinical condition characterised by apical retreat of the facial oral gingiva
- **Aetiology:** (Primary factor) morphology and the anatomy
 - Facial plate of bone overlying the root is usually very thin
 - Dehiscence or fenestrations are noted
- Anterior and premolars are most affected
 - Improper, traumatic tooth brushing – horizontal scrubbing, excessive force
 - Frenum pulls
 - Orthodontic treatment – tooth movement labially, arch expansion
- Radiographic presentation - true classical gingival recession localised to the facial surfaces cannot be seen

Classic Recession

- Interdental papilla fill entire embrasure area in young patients
- May be localised to one or present on several teeth
- Periodontal supporting structures are usually healthy
 - Teeth are not excessively mobile
 - No tooth loss due to classical gingival recession alone
- If OH is inadequate, it may lead to secondary inflammation and gingival and periodontal diseases

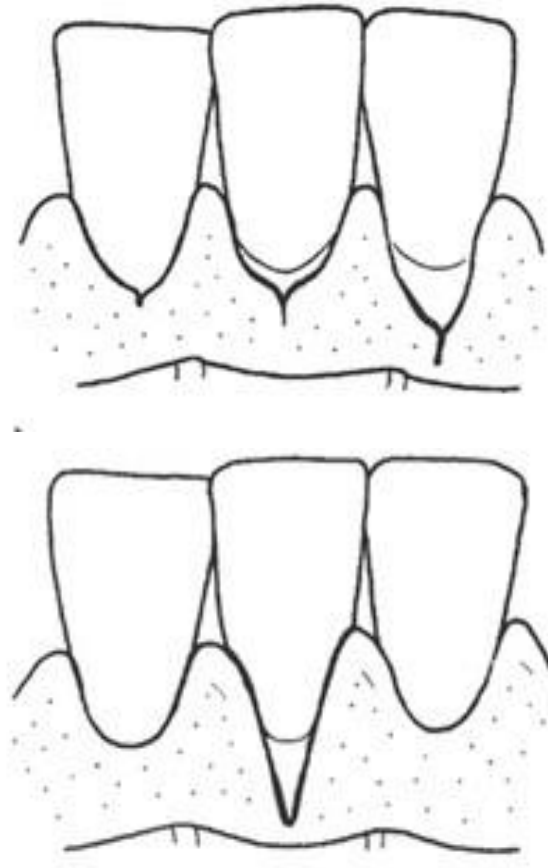


Gingival Recession and Types

- **Classic Recession:** apical retreat of gingiva with exposed root surface.
- **Stillman's Cleft:** narrow fissure or slit extending apically.
- **McCall's Festoon:** rolled, thickened marginal gingiva post-recession.
- **Dehiscence:** partial loss of alveolar bone covering the root.
- **Fenestration:** window of bone loss exposing root but leaving coronal bone intact.



Stillman's Cleft



Lippincott, Williams and Wilkins 2013



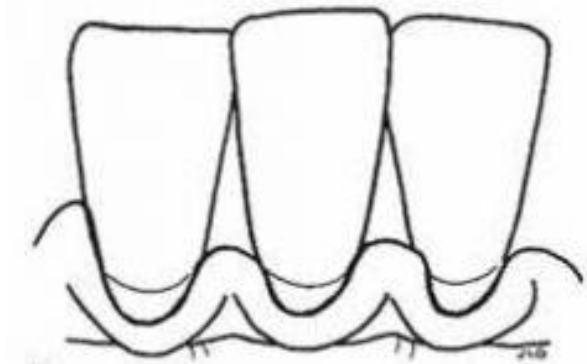
<http://www.rdpholistic.co.uk/images/gallery/case15before.jpg>



McCall's Festoon



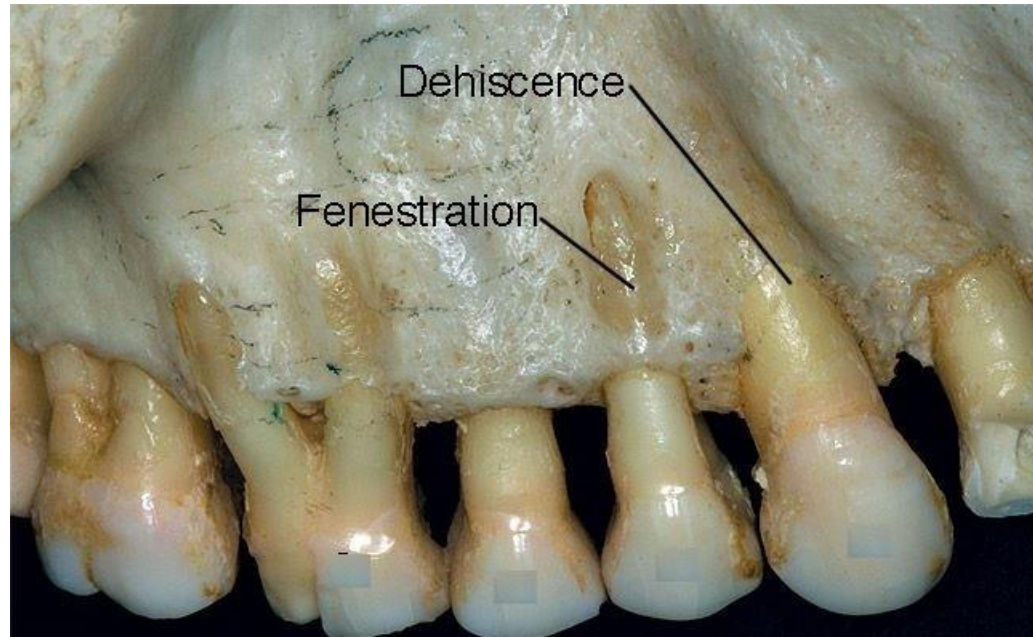
http://worcesterdentalimplants.com/_ah/editor/images/DSC_0258.JPG



Lippincott, Williams and Wilkins
2013



Dehiscence / Fenestration



<http://www.juniordentist.com/wp-content/uploads/2012/09/Dehiscence-and-Fenestration.jpg>

Dehiscence: is the incomplete coverage by bone over an area of a root that includes the cementoenamel junction

Fenestration: is a window of bone loss that exposes the root surface to the gingival or alveolar mucosa. The fenestration is bordered by alveolar bone on the coronal surface



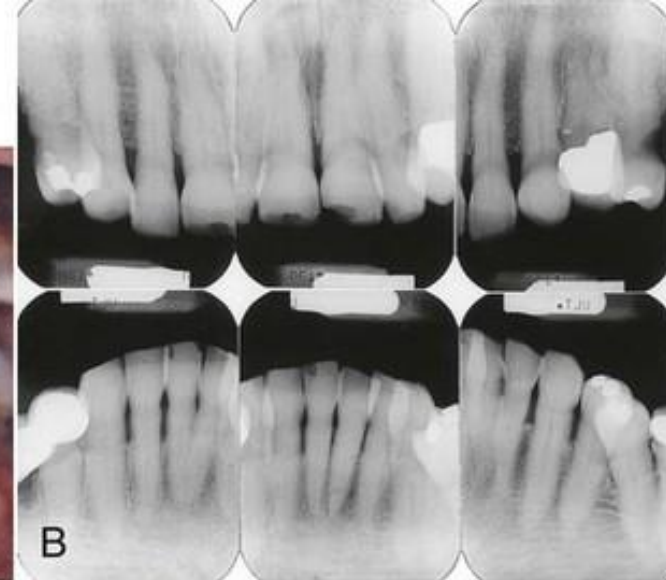
Recession- Diagnosis

- **Determine whether a recession is due to:**
 - Classical recession without inflammation
 - Pocket formation - periodontitis
 - Treated or untreated periodontitis
- **Additional significance due to increased risks for:**
 - Cervical abrasion
 - Dentinal sensitivity (less of a problem) - increased secondary dentin
 - Predisposition to root caries



Treatment for recession you may see...





<http://pocketdentistry.com/chapter-2-aging-and-the-periodontium/>



Brännström's Hydrodynamic Theory



Brännström's Hydrodynamic Theory

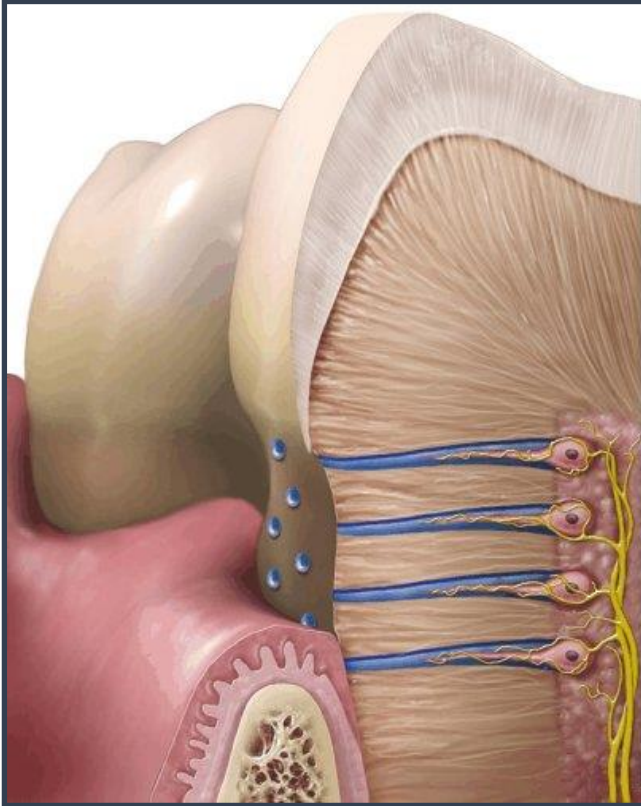


Diagram from:
<http://www.dentist-charlotte-north-carolina-nc.com/images/root-canal-dentinal-tubules-3-anatomy.jpg>

- Most widely accepted theory
- A stimulus (cold, heat or blast of air) on exposed dentin causes rapid fluid movement within dentinal tubules
- Fluid movement displaces or disturbs pulpal nerve endings causing pain



Brännström's hydrodynamic theory

- Dentinal tubules extend the thickness of the dentin
- **Odontoblastic processes** present in tubules (thin tail of cytoplasm from odontoblast in tooth pulp)
- Disturbances to the surface of exposed dentin where tubules are open result in fluid movement
- Fluid movement stimulates certain nerve endings associated with the odontoblastic processes causing **short, sharp pain** in the tooth

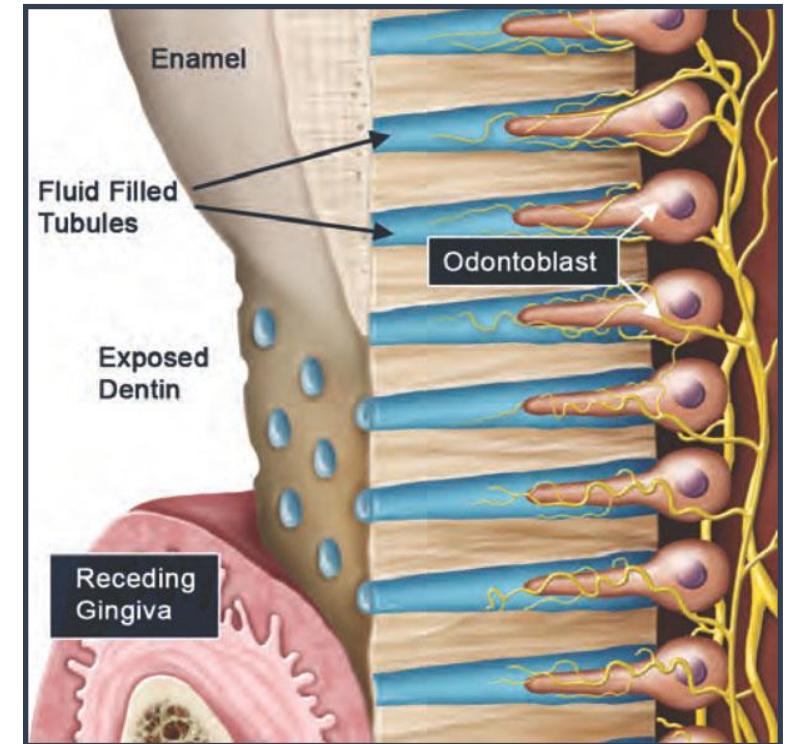
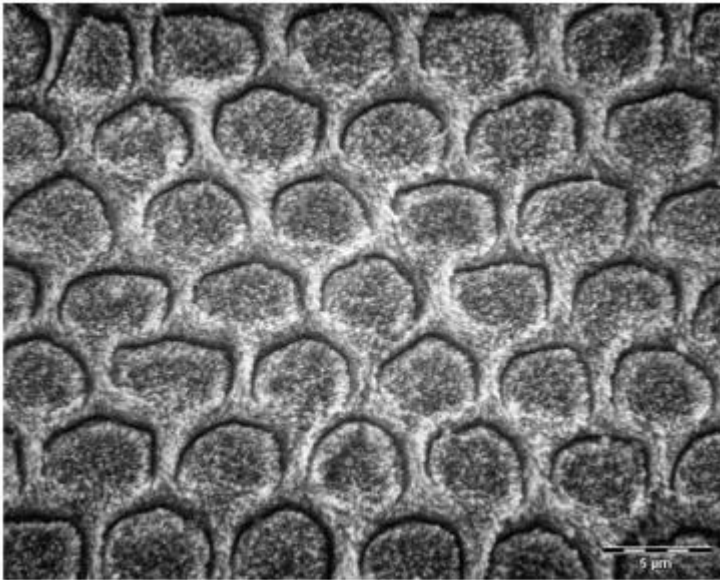


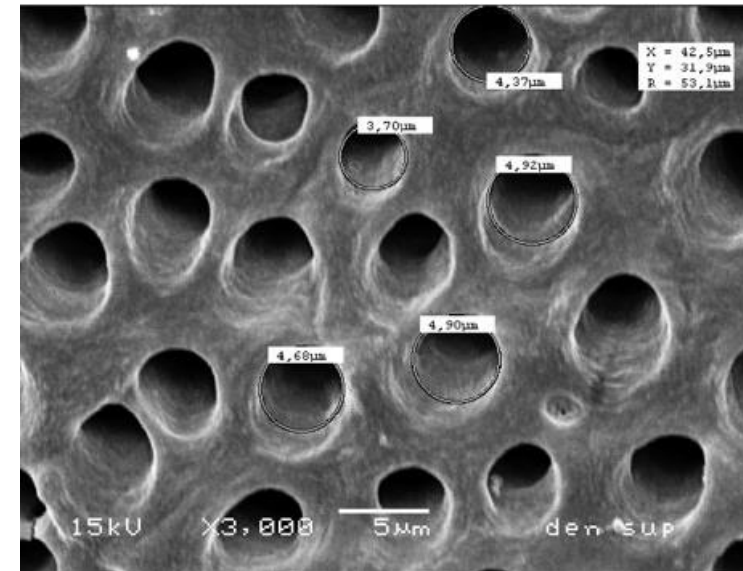
Image from:
<http://www.speareducation.com/spear-review/wp-content/uploads/2012/07/Dentinal-tubules-image-copy.jpg>

Microscopic Characteristics



ENAMEL

Image from:
<http://www.paulweberdds.com/learning-center/the-hard-tissues-of-the-tooth/the-enamel/>



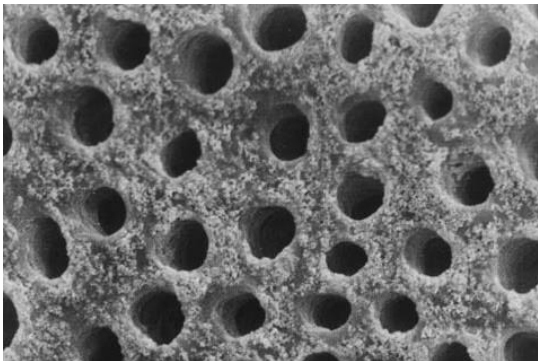
DENTINE

Image from:
Lopes, M. B., Sinhoreti, M. A., Gonini Júnior, A., Consani, S., & McCabe, J. F. (2009). Comparative study of tubular diameter and quantity for human and bovine dentin at different depths. *Brazilian Dental Journal*, 20, 279-283.

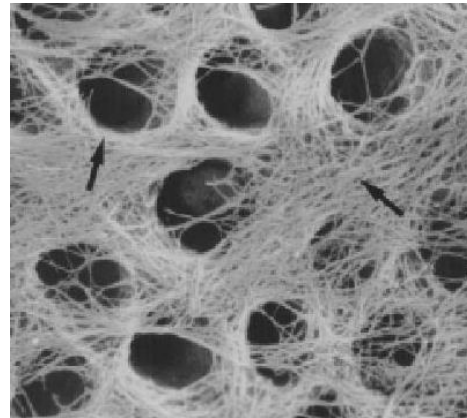


Microscopic Characteristics

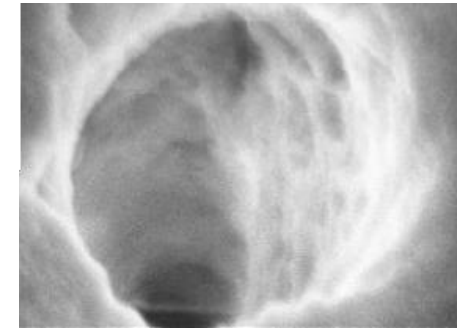
- More open & unobstructed dentinal tubules
 - 8 times more open dentinal tubules in sensitive dentin than in non-sensitive dentin
- Wider diameter of dentinal tubules
 - diameter of open tubules in sensitive dentin is twice that of non-sensitive dentin



Dentinal tubules (2000X magnification)



Dentinal tubules: Electron microscope (5000X)



Inner wall of dentinal tubule



Types of Pain Stimuli

Mechanical:

- Toothbrush bristles
- Eating utensils
- Dental & periodontal instruments
- Friction from denture clasps or other appliances
- Oral habits



Types of Pain Stimuli

Chemical:

- Acids in different forms
- Excessive intake of carbonated beverages
- Bacterial plaque acid production
- Citrus fruit acid
- Wine
- Spices
- Stomach acids (assoc. with gastric reflux, morning sickness, etc.)



Types of Pain Stimuli

Thermal:

- Hot or cold foods/beverages
- Air

Osmotic:

- Concentrated salt or sugar solutions



Conditions that Mimic Hypersensitivity

- Cracked tooth syndrome
- Defective restorations & marginal leakage
- Fractured or chipped teeth or restorations
- Dental caries
- Sensitivity following restorative treatment
- Incomplete polymerisation of restorative materials



Diagnosis of Hypersensitivity

Established:

- Thorough patient history
- Radiographic evaluation
- Clinical examination
- Specific questioning
- Clinical application of air, cold or mechanical exploration for isolation
- **Rule out other possible causes**





Strategies for Managing Hypersensitivity



Managing Hypersensitivity

- Instrumenting areas with existing hypersensitivity may result in sharp pain
 - Local anesthesia may be necessary for patient comfort
- Most instrumentation of root surfaces does not cause dental hypersensitivity
- Most areas of hypersensitivity eventually desensitise on their own (**smear layer**)
- Dentinal tubules go through a natural process of crystallisation and occlusion (blocking of the open dentinal tubules)
 - Natural process takes several weeks



Managing Hypersensitivity

- Strategies for intervention and management can include:
 - **Chemical Management (Chemotherapeutic Agents or Preventive Tx)**
 - **Patient Education**

Patients undergoing non-surgical periodontal therapy MUST be warned about the possibility of hypersensitivity before initiating any treatment.



Smear Layer

- **Crystalline debris (naturally occurring)** from the tooth surface that plugs the dentinal tubules
- Inhibits fluid movement, thus preventing sensitivity
- Can be removed by instrumentation
- Irritation and stimulation encourage secondary dentin formation

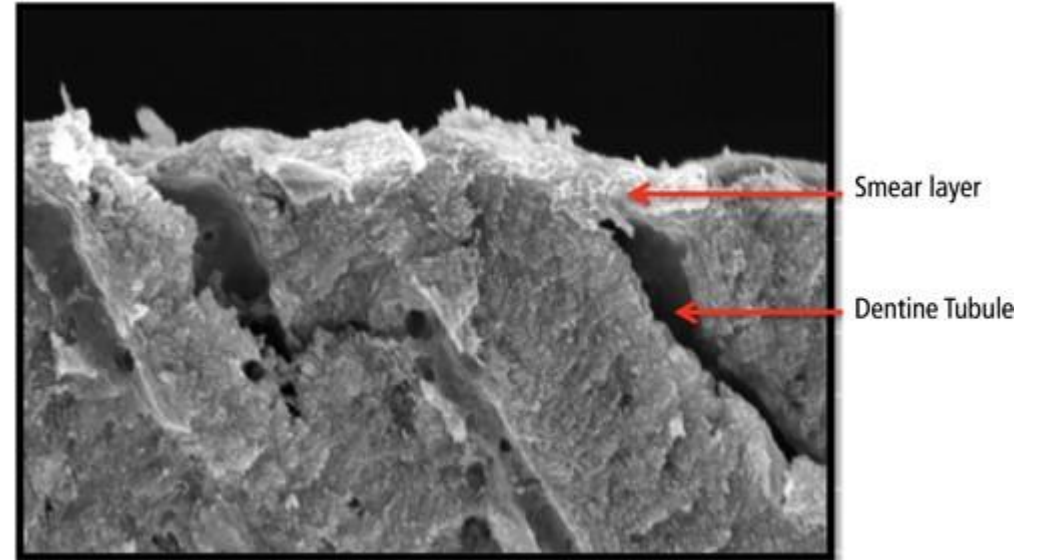


Image from:
Olley, R. C., & Sehmi, H. (2017). The rise of dentine hypersensitivity and tooth wear in an ageing population. *British Dental Journal*, 223(4), 293-297..



Spontaneous Remission

- **Smear layer formation**
 - Deposition of fluoride and minerals from diet, drinking water, dentifrice & minerals in saliva
- **Tubular sclerosis (narrowing of tubules)**
 - Can occur naturally as root dentin ages but can also be caused by trauma, abrasion, or bacterial invasion
- **Secondary dentin**
 - Irritation and stimulation encourage secondary dentin formation, retracting odontoblasts
- **Formation of calculus**



Desensitising Agents Ingredients

- Blocks the dentinal tubule openings
- Interrupts nerve transmission

No single therapy is proven to be effective in all situations or for all patients.

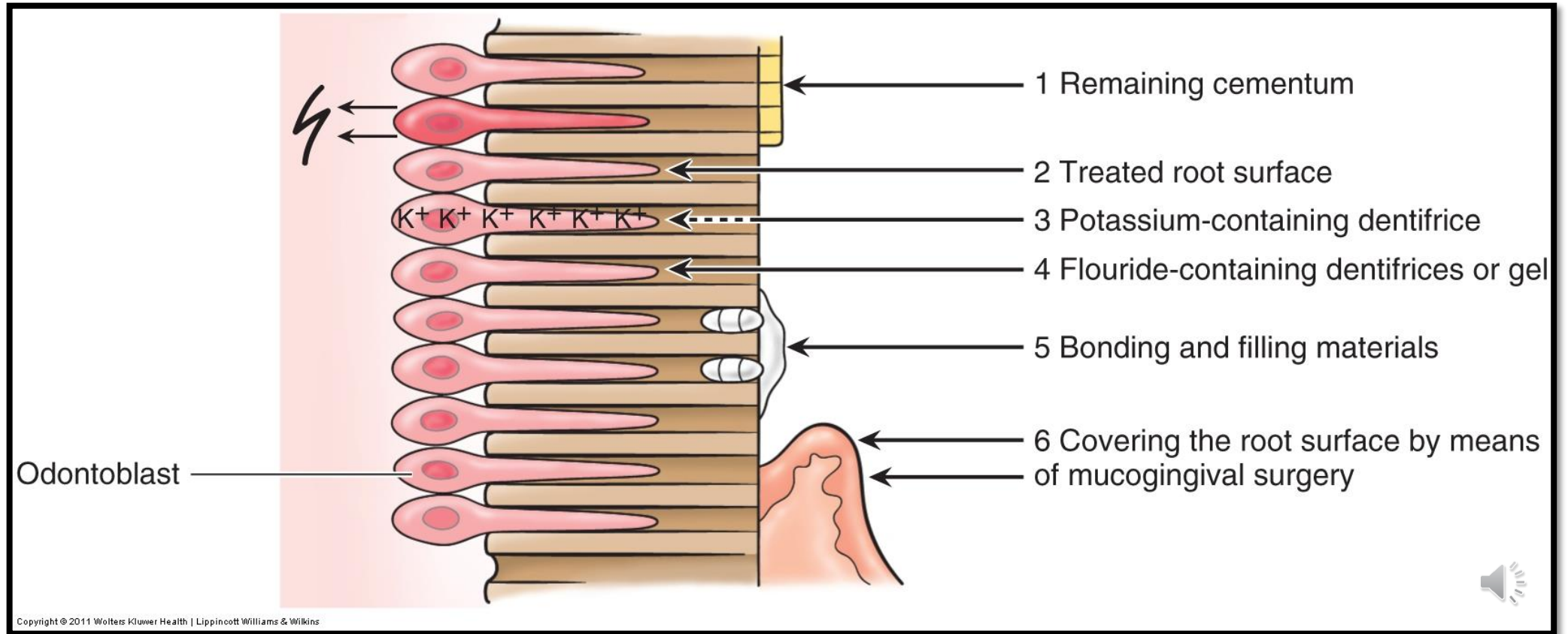


Desensitising Agents Ingredients

- **Potassium** nitrate (Sensodyne)
- Stannous **fluoride** (Sensodyne)
- Sodium **fluoride** (Duraphat)
- Arginine – (Amino acid - Colgate Sens Pro Relief)
- Laser
- Sodium citrate
- Strontium chloride
- Glutaraldehyde with HEMA
- Glass Ionomer cements
- Hydroxyapatite
- Adhesive systems
- Silver Fluoride (Riva star)
- Potassium Iodide (Riva star)



Dentinal Tubules After Different Types of Therapy



Home-Care Management

Toothpastes for desensitisation:

- Potassium nitrate / chloride / citrate → nerve depolarisation.
- Strontium chloride / acetate → tubule occlusion.
- Fluoride (NaF, SnF₂) → form calcium fluoride layer.
- Arginine + calcium carbonate (Pro-Argin™) → occlusion.
- CPP-ACP (Tooth Mousse Plus) → remineralisation.



Professional / In-Office Management

- Fluoride varnishes: Duraphat, Clinpro, Enamel Pro.
- Desensitising pastes: Pro-Relief (Arginine), oxalate sealants.
- Bonding agents or resin sealants: occlude tubules.
- Periodontal surgery: soft tissue graft for recession.
- Endodontic therapy: in severe non-responsive cases.



Mechanism of Common Agents

Agent	Mechanism	Example
Potassium nitrate	Nerve depolarisation	Sensodyne Daily Care
Strontium chloride	Tubule occlusion	Macleans Sensitive
Fluoride varnish	Tubule sealing	Duraphat
Arginine + CaCO_3	Calcium plug formation	Colgate Pro-Relief
CPP-ACP	Remineralisation	Tooth Mousse Plus



Post-Periodontal Treatment Sensitivity

- Common occurrence following:
 - Scaling and root planing (SRP)
 - Non-surgical debridement
 - Periodontal surgery
- Why it happens:
 - Removal of calculus and cementum exposes dentinal tubules.
 - Removal of the smear layer increases dentine permeability.
 - Gingival shrinkage post-therapy exposes root surfaces.
 - Temporary pulpal inflammation enhances nerve excitability.



Natural Recovery Mechanisms

- Formation of a new smear layer from salivary minerals.
- Tubular sclerosis and secondary dentine formation.
- Mineral precipitation from fluoride and calcium.
- Gradual reduction of fluid movement and pain.



Patient Education

- Inform patients pre-treatment about potential sensitivity.
- Reassure that sensitivity is transient and usually resolves within weeks.
- Recommend gentle brushing with soft-bristle toothbrush and desensitising toothpaste.
- Advise avoiding cold drinks or acidic foods for 48 hours post-treatment.

